Assignment 3: Data Science for Software Engineers
Exploratory Data Analysis on Air Quality Data

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1 Introduction

This assignment involves performing exploratory data analysis (EDA) on a dataset containing air quality measurements collected over a year. The aim is to identify trends, patterns, and the relationship between pollutants and weather variables.

2 Section 1: Data Cleaning

2.1 Identifying Issues

The dataset contains some missing values, outliers, and inconsistencies. For example, negative pollutant values and unrealistic weather data such as extremely low or high temperatures were observed.

2.2 Handling Missing Values and Outliers

Missing values were handled using **interpolation** for continuous variables and mode imputation for categorical data. Outliers were handled using the **Interquartile Range (IQR)** method, and extreme values were clipped to preserve data continuity.

3 Section 2: Feature Engineering

3.1 Air Quality Index (AQI)

A new column **air_quality_index** was calculated using:

$$AQI = \frac{PM2.5 + PM10 + NO2 + O3}{4}$$

3.2 Pollution Level Categories

A new column **pollution_level** was created:

- Good: AQI ≤ 50
- Moderate: $50 <$ AQI ≤ 100
- Unhealthy: AQI > 100

3.3 Season Column

A new column **season** categorizes months:

- Winter: November , December, January, February
- Spring: March, April, May
- Summer: June, July, August
- Fall: September, October

3.4 Weekday/Weekend Column

A column **weekday** categorizes days as **Weekday** or **Weekend** based on the day of the week.

4 Section 3: Visualization

4.1 Time-Series Plot of AQI

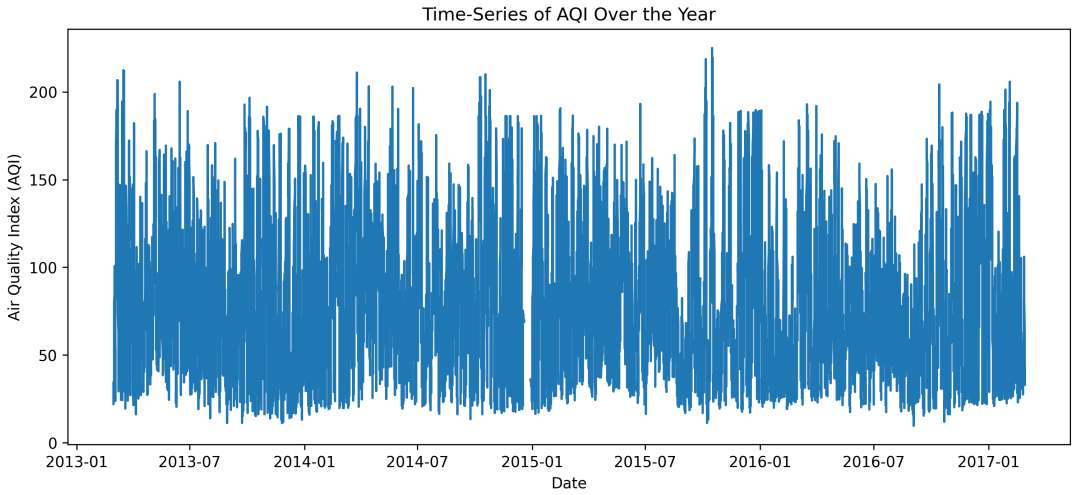

Figure 1: Time-series plot of air quality index over the year.

Insights: Seasonal patterns are visible.
Higher AQI values are observed in winter months,
indicating poor air quality during this period.

4.2 Pollutant Comparison Across Seasons

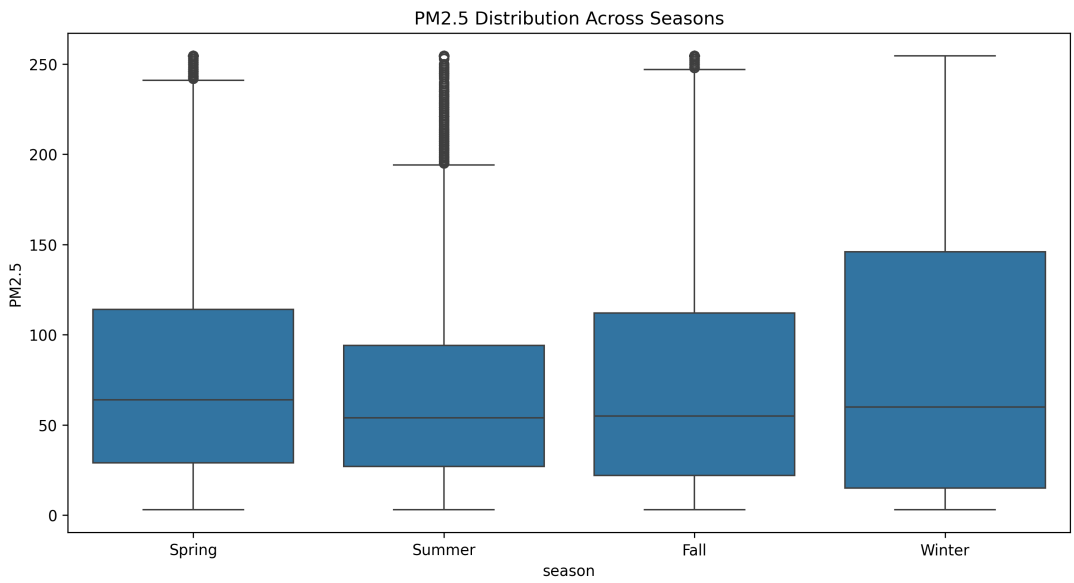

Figure 2: Distribution of PM2.5 across seasons

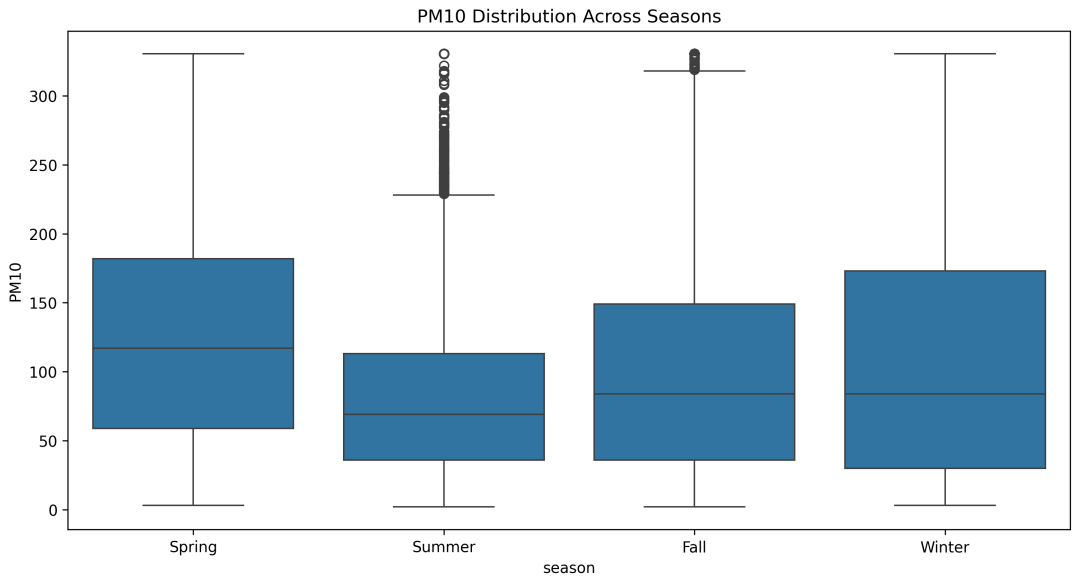

Figure 3: Distribution of PM10 across seasons

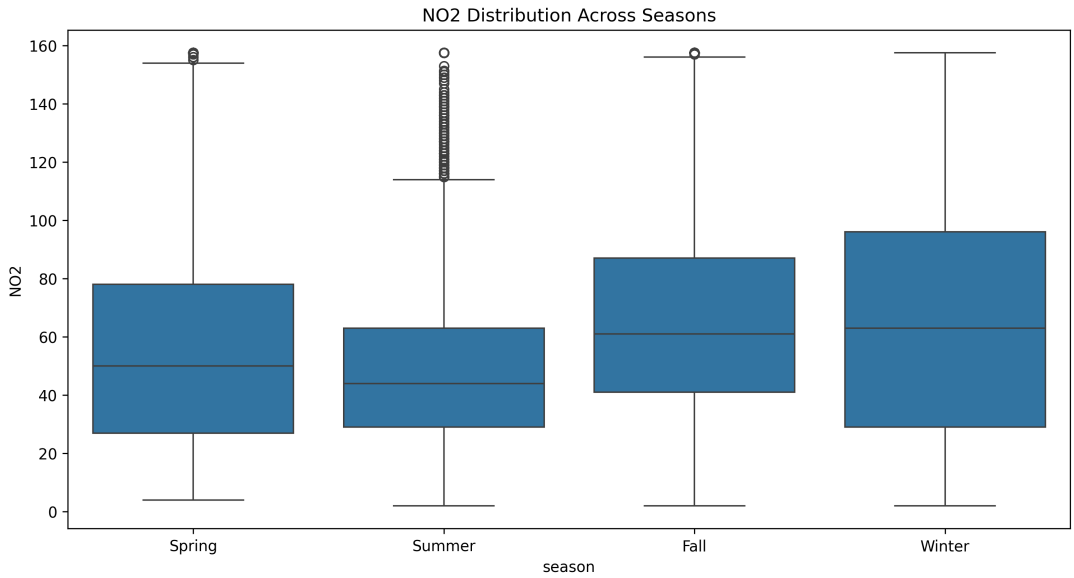

Figure 4: Distribution of NO2 across seasons

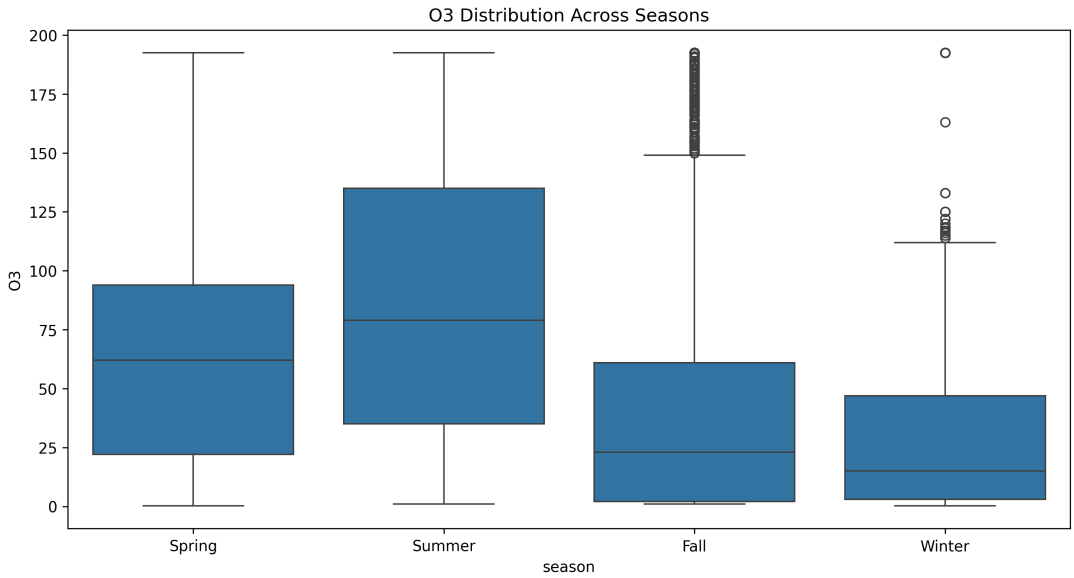

Figure 5: Distribution of O3 across seasons

Insights: Boxplots show how each pollutant varies in each season.

If winter boxes are higher (winter pollution is worse).

If monsoon has low values (rainy season cleans the air).

More outliers (more unstable pollution levels) .

Pollution is generally higher in winter.

PM2.5 and PM10 show more variability in winter, while O3 levels are higher in summer.

4.3 Temperature vs AQI Scatter Plot

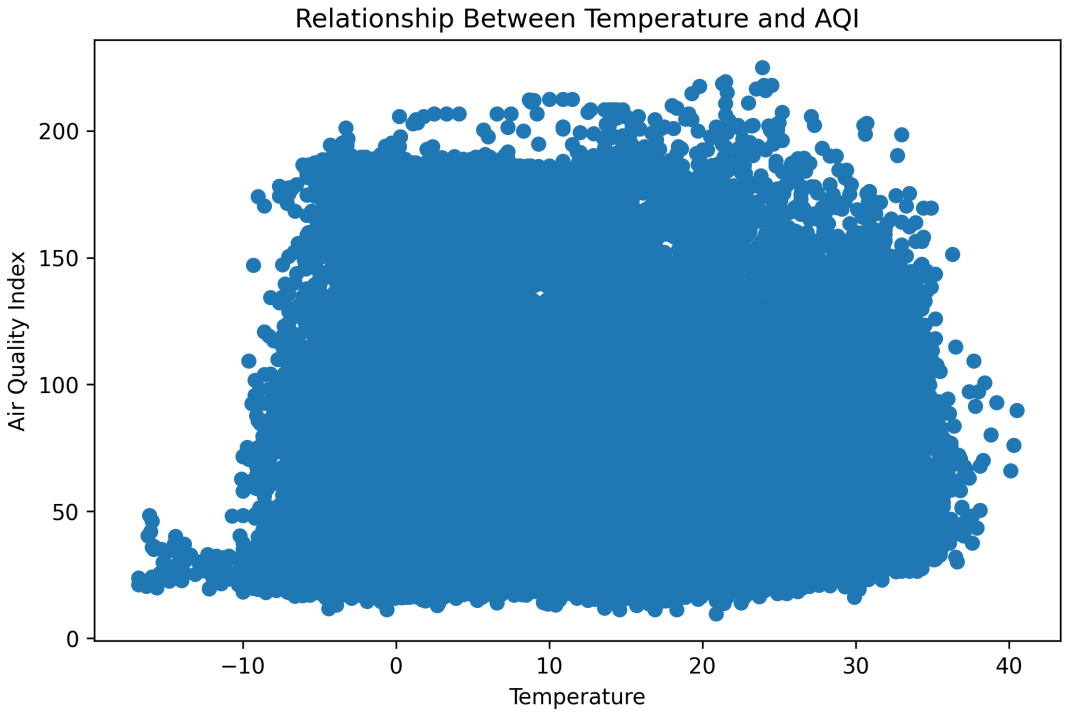

Figure 6: Scatter plot showing relationship between temperature and AQI.

Insights: Points going upward (higher temperature associated with higher pollution).
Points going downward (clean air in warmer weather).
No pattern (no strong relationship).

4.4 Pollution Level Analysis

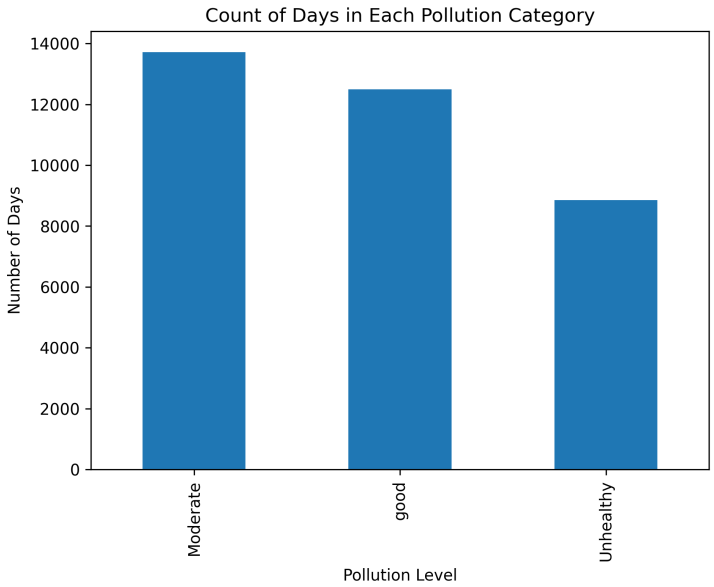

Figure 7: Count of days in each pollution category.

Insights: Shows which category is most common.
If "Moderate" is highest (general pollution but not severe daily).
If "Unhealthy" has many days (public health concern).

4.5 Monthly AQI Distribution

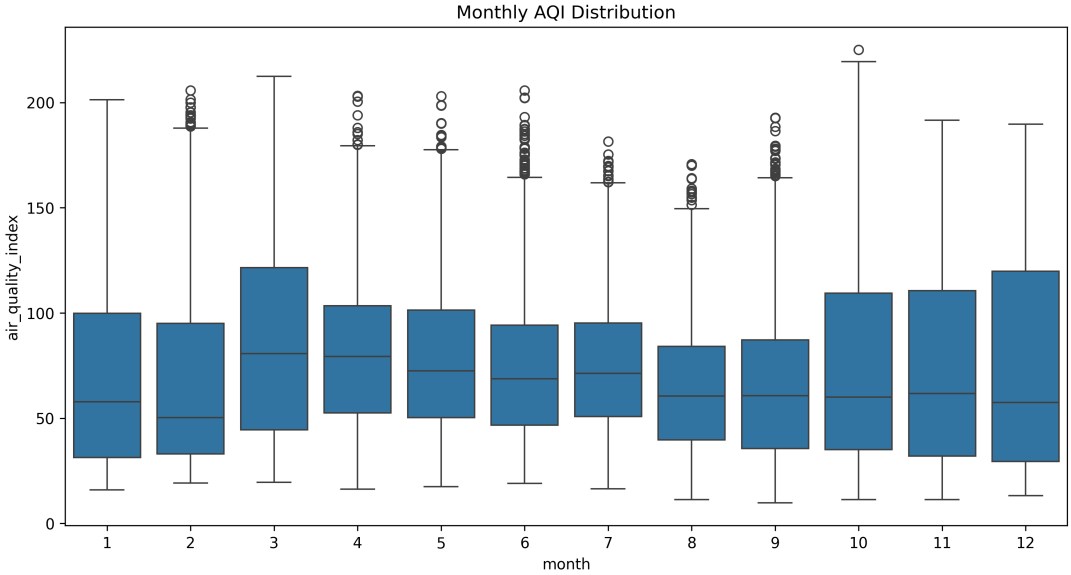

Figure 8: Monthly AQI distribution for the year.

Insights: Shows which months have high/low AQI.
A rising or falling pattern (seasonal effect).
Anomalies (very high values in unexpected months), special pollution events.

5 Conclusion

EDA revealed clear seasonal and weather-related patterns in air quality. The feature engineering helped summarize air pollution trends, and the visualizations provided insights into pollutant distributions, seasonal effects, and the relationship between weather and air quality.